



Batterjee Medical College
For Science and Technology

Thyroid Disorders

Internal Medicine Department

Knowledge

- Relate pathophysiological basics and aetiology of hypothyroidism and other thyroid disorders
- Describe epidemiology, manifestations, complications and management of hypothyroidism

Skills

- Interpret clinical manifestations and investigations of hypothyroidism and other thyroid disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for hypothyroidism and other thyroid disorders
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

- Hypothyroidism (goitrous and non-goitrous causes, clinical manifestations across systems, myxoedema coma, treatment)
- Thyroiditis (subacute thyroiditis: investigations and treatment)
- Thyroid carcinoma (papillary, follicular, anaplastic, medullary, lymphoma: frequency, behaviour, spread, treatment)
- Subclinical thyroid dysfunction (subclinical hypothyroidism and hyperthyroidism: definition, symptoms, complications, management)
- Diagnostic approach to thyrotoxicosis by aetiology (Graves' disease, toxic multinodular goiter/solitary nodule, thyroiditis, iodine-induced, and other causes)



HYPOTHYROIDISM

Causes :

**1- 1ry
hypothyroidism
(95%).**

**2- 2ry
hypothyroidism
(5%) due to
hypothalamic or
pituitary disease.**

**3- Tissue resistance
to thyroid hormone
(rare).**

Non goitrous

- Congenital developmental defect
- Post-surgery
- Post I131 therapy;
- idiopathic
 - It may be associated with antithyroid antibodies or antibodies to TSH receptors.
 - May be associated with auto antibodies to other endocrinal glands.
 - May be associated with other autoimmune disease e.g. SLE, myasthenia gravis.

Goitrous

- Inherited biosynthetic defect (+ congenital deafness) → pendred syndrome.
- Iodine deficiency → endemic goitre.
- Maternally transmitted if antithyroid drug or I131 during pregnancy)
- Drug induced :
 - Phenyl butazone.
 - PAS.
 - Antithyroid drugs.
- Chronic thyroiditis: Hashimoto's disease, Riedel's thyroiditis
- Self limited :
 - Subacute thyroiditis.
 - Postpartum thyroididtis

- While myxedema is often associated with severe hypothyroidism, it's not entirely the same. Myxedema specifically refers to the skin and tissue swelling caused by the accumulation of mucopolysaccharides (a type of carbohydrate) in hypothyroidism. It's a symptom, rather than a disease itself. Hypothyroidism is a broader condition where the thyroid gland doesn't produce enough thyroid hormone.
- It is much common in ♀ than ♂ (usually between 30 - 50 years), mostly auto-immune process (circulating thyroid antibodies in > 80 % of cases).

Clinical picture :

1.Neurological manifestations :

- Reduced memory, mental slowing, dementia may occur → Myxoedema Madness, depression.
- Delayed relaxation of tendon jerks (suspended jerks).
- Mucinous infiltration:
 - o In the flexor retinaculum → carpal tunnel syndrome.
 - o In the internal ear → progressive deafness.
 - o In the Vocal cords → hoarsness of voice.
 - o In the tongue → slurred speech.

2.Cardiovascular manifestations:(+ hematological manifestation)

- Sinus bradycardia.
- Cardiomyopathy → heart failure.
- Cholesterol pericarditis & pericardial effusion.
- Atherosclerosis → angina pectoris & intermittent claudication.
- Hypertension: due to ↑ peripheral resistance.
- Anemia
 - o Normocytic: due to → bone marrow depression & ↓ peripheral O₂ requirements.
 - o Megaloblastic: due to associated pernicious anemia.
 - o Microcytic hypochromic :due to → menorrhagia & achlorhydria.

3.Pulmonary manifestations :

- Pleural effusion. '
- ↓ Ventilatory response to hypoxia & hypercapnia → CO₂ retention.

4.Renal manifestations:

- ↓ excretion of a water load → hyponatraemia (may be due to SIADH).

5.GIT manifestations :

- Adynamic ileus:
 - Constipation.
 - Intestinal obstruction.
- Achlorhydria: often associated with pernicious anaemia.
- Ascites (with high cholesterol content).

6. Musculoskeletal manifestations:

- Arthralgia, joint effusion & stiff muscles

7. Skin " & hair " manifestations:

- Puffy face & coarse features

➤ Skin :

- Dry cold skin.
- Orange due to accumulation of carotene (carotene → thyroxin → vit A).
- Malar flush.

➤ Hair:

- Sparse, brittle, coarse. .
- Loss of hair from the outer 1/3 of the eye brows.

➤ Xanthelasma (due to ↑ cholesterol).





8.Reproductive manifestations : .

- Menorrhagia 'due to anovulatory cycles' → the commonest.
- Amenorrhea & galactorrhea in cases associated with hyperprolactinaemia (late).

9.Metabolic & endocrinal manifestations :

- Growth & development retardation in children (Juvenile myxedema).
- Growth hormone deficiency (because thyroxin is necessary for GH synthesis).
- Gain of weight (despite ↓ appetite).
- Hypothermia & cold intolerance.
- Hyperlipidaemia (due to ↓ lipoprotein degradation).

10. Thyroid gland : may be:

- Atrophic.
- Enlarged.
- Show evidence of previous thyroidectomy.

HYPOTHYROIDISM



11. Myxoedema coma :

➤ Causes

- Old standing untreated hypothyroidism.
- Hypothyroidism with exposure to :
 - o coldness
 - o Infection.
 - o Trauma.
 - o CNS depressants.

➤ Clinical picture :H

- External features of severe hypothyroidism & bradycardia.
- Hypothermia.
- Dilutional hyponatraemia.
- Alveolar hypoventilation → CO₂ retention & narcosis.
- loss of consciousness.
- Hypoglycaemia.
- Heart failure & Hypotension

Investigations:

1. Evaluation of thyroid function.
2. X-ray chest & heart → pericardral effusion or pleural effusion.
3. ECG → low voltage, bradycardia.

Differential diagnosis:

1. 1ry & 2ry hypothyroidism
2. Nephrotic syndrome :

- Similar in myxoedema in :
 - o Facial- puffiness & pallor.
 - o Hypercholesterolemia.
 - o ↓total T4, T3 (due to loss of TBG in urine).

- Different in :
 - o Normal TSH.
 - o Heavy proteinuria.

3. Chronic renal failure :

- Similar to myxoedema in :
 - o Facial puffiness & pallor.
 - o hypertension.
 - o Dry skin.

- Different in :
 - o Polyuria, pruritis.
 - o Normal thyroid function.

Treatment:

A. Preparation :

- Na Levo-thyroxin (L-thyroxin) → of choice.
- Desiccated thyroid extract (T4, T3: 4 : 1)
- Synthetic T3 (there is no real indication for its use because T4 is converted to T3 in the body).

B. Rapid correction in :

- Neonatal, infantile & Juvenile hypothyroidism.
- Myxedema coma.
- Hypothyroid patient prepared to emergency surgery → IV. administration of L-thyroxin + hydrocortisone is indicated.

C. Slow correction :

- In elderly patients or patients with heart disease start with 25 - 50 ug / day & ↑ the dose by 25 - 50 ug every month (according to the clinical response) until reaching 150 - 200 ug(full replacement dose)

D. Clinical response:

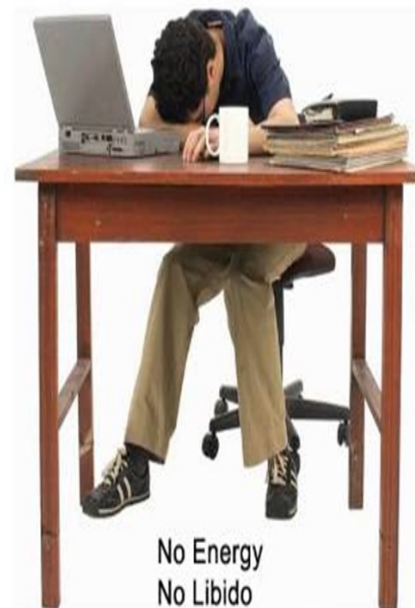
Full effect is likely to take 2 - 3 months.

E. monitoring the replacement dose :

- T4 becomes normal within few days.
- T3 (better) in 2 - 4 weeks.
- TSH in 6 - 8 weeks.

➤ treatment of myxoedema coma :

- L-thyroxin : 500 ug I.V.
- hydrocortisone 100 mg(for possible associated adrenal insufficiency).
- Hypertonic saline (+ glucose).
- Assisted ventilation.
- Avoid further heat loss.



No Energy
No Libido
Cold sensitivity
Weight Gain
Mental Fog
Depression
Dry skin and hair
Goitre
(Menstrual changes)



No Sleep
Libertarian
Hot stuff
Rake thin
Mental Fire
Anxiety
Heart racing & tremor
Light periods

Findings	Hypothyroidism	Hyperthyroidism
Metabolic	Intolerance to cold Weight gain Decreased appetite	Intolerance to heat Weight loss Increased appetite
Neuropsychiatric	Excessive fatigue	Irritability, restlessness
Periorbital findings	Periorbital edema (due to generalized myxedema)	Periorbital edema, lid lag, and exophthalmos (in Graves ophthalmopathy)
Cardiovascular	Bradycardia, decreased cardiac output	Tachycardia, palpitations, hypertension
Skin	Cold, dry	Warm, moist
Gastrointestinal	Constipation	Hyperdefecation
Musculoskeletal	Cramps	Osteopathy
Edema	Generalized myxedema (initially pretibial)	Pretibial myxedema (in Graves disease)
Myopathy	Proximal	
Reproduction	♀ Menstrual disorders ♂ Decreased libido, infertility	
Hair	Hair loss	



THYROIDITIS

Causes :

- Usually follow an upper respiratory tract infection & can be due to any of several Viruses.

Clinical picture :

- Stretching of the thyroid capsule causes pain over the thyroid or referred pain to the lower Jaw, ear or occiput. May be with fever & nodularity of the thyroid.
- Dysphagia.

Investigations :

- ↑ T4 (due leakage from the gland);after the hormone is depleted an euthyroid or hypothyroid phase occurs, eventually return to normal (recovery Phase) .
- ↓ RAIU , ↑ ESR.

Treatment :

- Mild cases → aspirin to control symptoms.
- Severe cases :
 - Prednisone 15-20 mg tid
 - Propranolol 'to control thyrotoxic manifestation.

Cause:

- This chronic thyroid inflammation is autoimmune in origin & may coexist with other autoimmune diseases e.g. SLE, Pernicious anemia, Sjogren's syndrome.

Clinical picture:

- Goitre may be asymptomatic & thyroid failure usually occur
- Initially serum T4 $\uparrow$ then $\downarrow$ & frank hypothyroidism occur.
- High titre of thyroid antimicrosomal antibodies. Such antibodies may cause simultaneous thyroiditis & hyperthyroidism $\rightarrow$ hashitoxicosis.

Treatment:

Levothyroxin (in replacement doses) $\rightarrow$ regression of goiter.



Solitary Thyroid Nodules

- Thyroid nodules are common. They can be detected clinically in about 4% of the population and are found in about 50% of the population at autopsy.
- Patients may have one prominent nodule on clinical examination, but thyroid ultrasound evaluation may reveal multiple nodules.
- Although most nodules are benign, a small percentage are malignant.
- In addition, most thyroid cancers are low-grade malignancies.

- The major etiologic factor for thyroid cancer is childhood or adolescent exposure to head and neck radiation.
- Patients with a history of irradiation should have a baseline thyroid ultrasound and their thyroid carefully palpated every 1 to 2 years.
- A dominant nodule (>1 – 1.5 cm) or nodules with ultrasound features compatible with neoplasia should undergo FNA.
- Although in the past benign thyroid nodules were treated with levothyroxine suppression, this is no longer recommended because it is uncommon for thyroid nodules to shrink substantially with levothyroxine.

Table 66-6 High-Risk Factors for Malignancy in a Thyroid Nodule

History

Head and neck irradiation
Exposure to nuclear radiation
Rapid growth
Recent onset
Young age
Male sex
Familial incidence (medullary and about 5% of papillary)

Physical Examination

Hard consistency of nodule
Fixation of nodule
Lymphadenopathy
Vocal cord paralysis
Distant metastasis

Laboratory and Imaging

Elevated serum calcitonin
Cold nodule on technetium scan
Solid lesion with microcalcifications on ultrasonography

Thyroid Cancer



PAPILLARY

- ✦ Most common type of thyroid cancer
- ✦ Represents 75 - 85% of thyroid cases

75-85%



FOLLICULAR

- ✦ Well differentiated tumor
- ✦ Accounts for 15% of thyroid cases

15%



MEDULLARY

- ✦ Originates from the parafollicular cells
- ✦ Represents 3 % of the thyroid cases

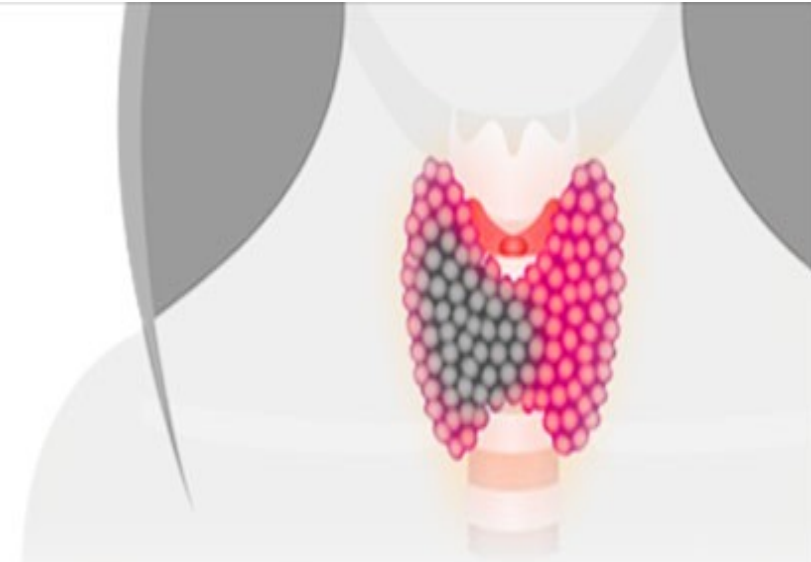
3%



ANAPLASTIC

- ✦ Most aggressive form of thyroid cancer
- ✦ Accounts for only 1% of the thyroid cases

1%



Cell type	Frequency	Behavior	Spread	Treatment
Papillary	70%	Commonest. May be multifocal. Occurs in young people (F>M)	- Spreads via <u>lymph nodes</u> . - Local LN metastases occurs even with occult tumors < 1cm.	Total thyroidectomy & adjuvant 131-Iodine. Thyroxin give if hypothyroid and to reduce recurrence. <u>Good prognosis</u>
Follicular	20%	Single encapsulated lesion. Mean age 50 yrs. (F>M)	Spreads via <u>blood</u> , to lung/bone,...	As above
Anaplastic	<5%	<u>Most aggressive</u> . Hard rapidly enlarging. Wt loss, stridor, hoarseness	Locally invasive	Surgery, chemo, radio - <u>useless</u> 80% die within 1 yr.
Lymphoma	2%	Variable		Sometimes responsive to radiotherapy
Medullary cell	5%	Arises from calcitonin secreting parafollicular C cells. Often familial & associated with MEN syndrome.	Lymphatic spread has frequently developed.	- Total thyroidectomy - 131-iodine has no role. - Poor prognosis, but indolent course

Subclinical hypo and hyper Thyroidism

Subclinical Hypothyroidism:

- **Def:** Elevated TSH (above normal range) with normal levels of T4 and T3.
- **Symptoms:** Typically asymptomatic, but may present with subtle symptoms like fatigue or constipation.
- **Potential complications:** May be associated with increased risk of cardiovascular disease, particularly in severe cases.

Subclinical Hyperthyroidism:

- **Def:** Low TSH (below normal range) with normal levels of T4 and T3.
- **Symptoms:** May include mild symptoms like anxiety, palpitations, or weight loss.
- **Potential complications:** May be associated with atrial fibrillation, bone loss, and increased risk of heart problems.

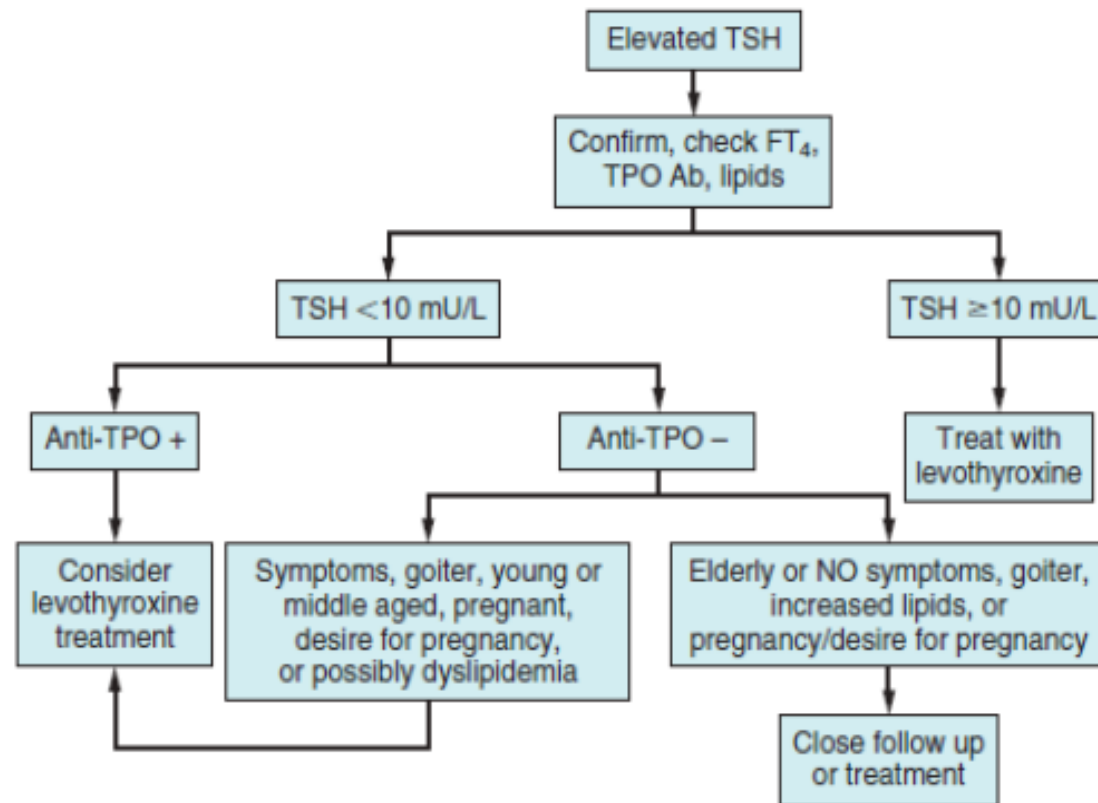


FIGURE 38-7 Management of subclinical hypothyroidism. Treatment is recommended for individuals with TSH 10 mU/L or greater. For individuals with TSH less than 10 mU/L, TPO antibody levels, age, pregnancy status, clinical symptoms/signs, and possibly lipids are important factors in considering whether to initiate treatment. FT₄, Free thyroxine; TSH, thyroid-stimulating hormone; TPO, thyroid peroxidase.

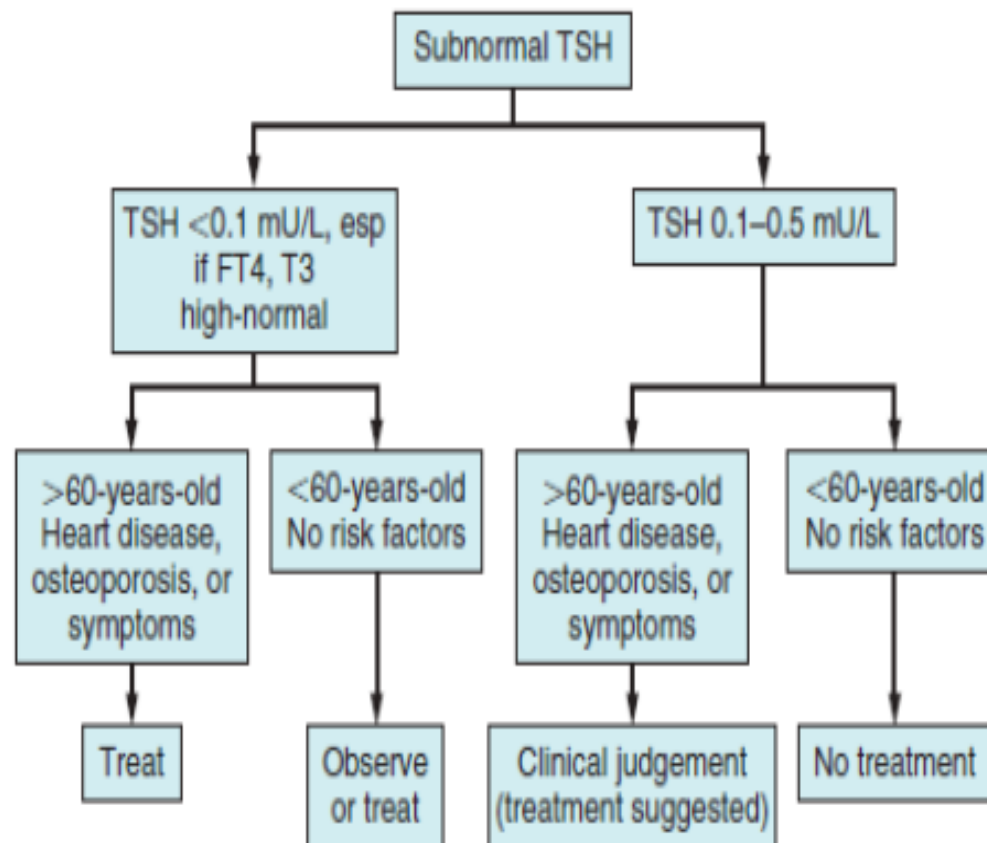


FIGURE 38-5 Management of subclinical hyperthyroidism. If the patient is older than 60 years or younger than 60 years with bone disease, heart disease, or symptoms, treatment is recommended. FT₄, Free thyroxine; TSH, thyroid-stimulating hormone.



Summary

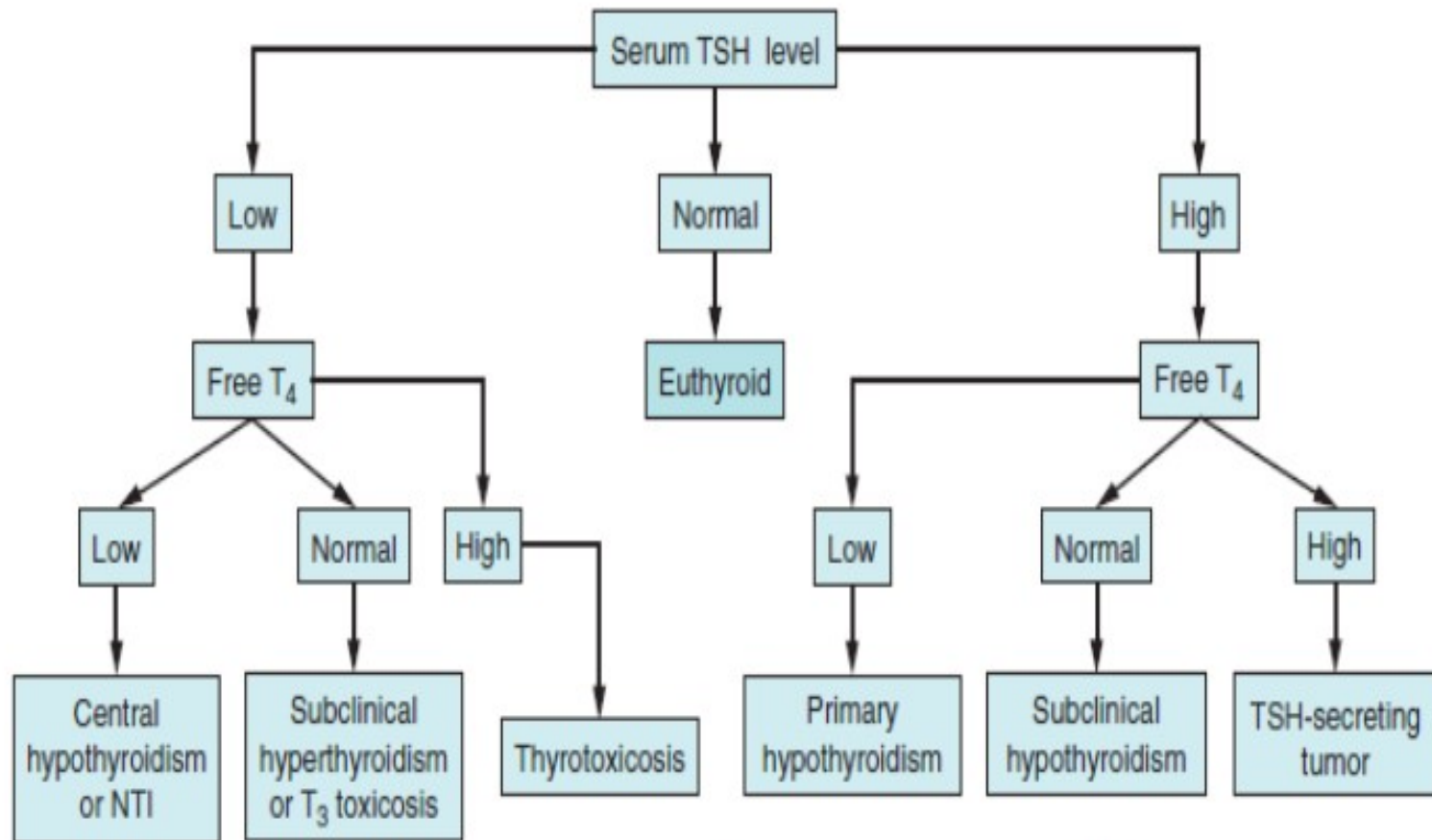
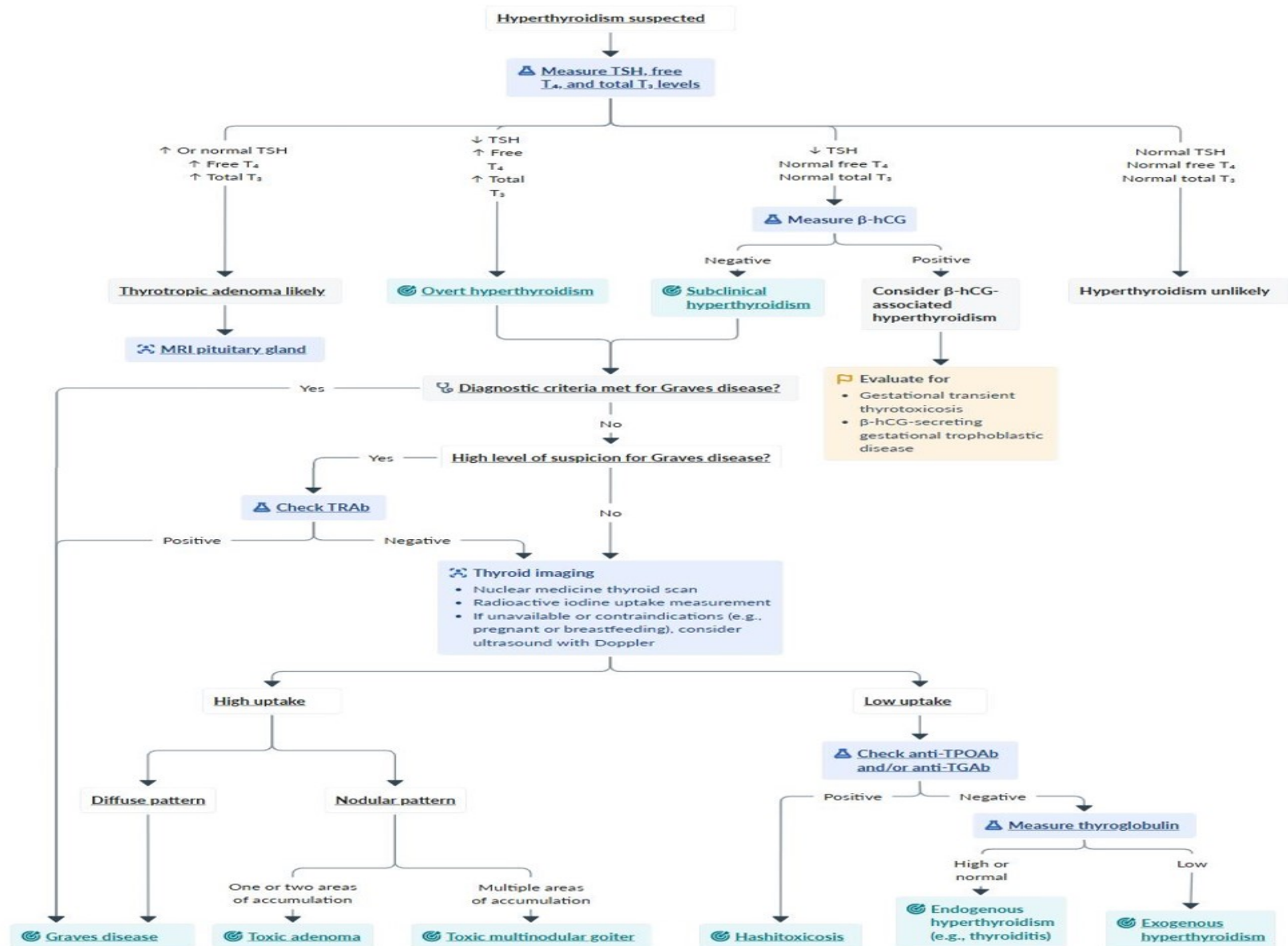


FIGURE 38-2 Sequential thyroid function testing using the serum thyroid-stimulating hormone (TSH) as a starting point. This strategy is useful for screening asymptomatic individuals. If symptoms are present, free thyroxine (FT₄) levels should also be assessed. NTI, Nonthyroidal illness; T₃, triiodothyronine.

Diagnosis of hyperthyroidism



Diagnosis of hypothyroidism

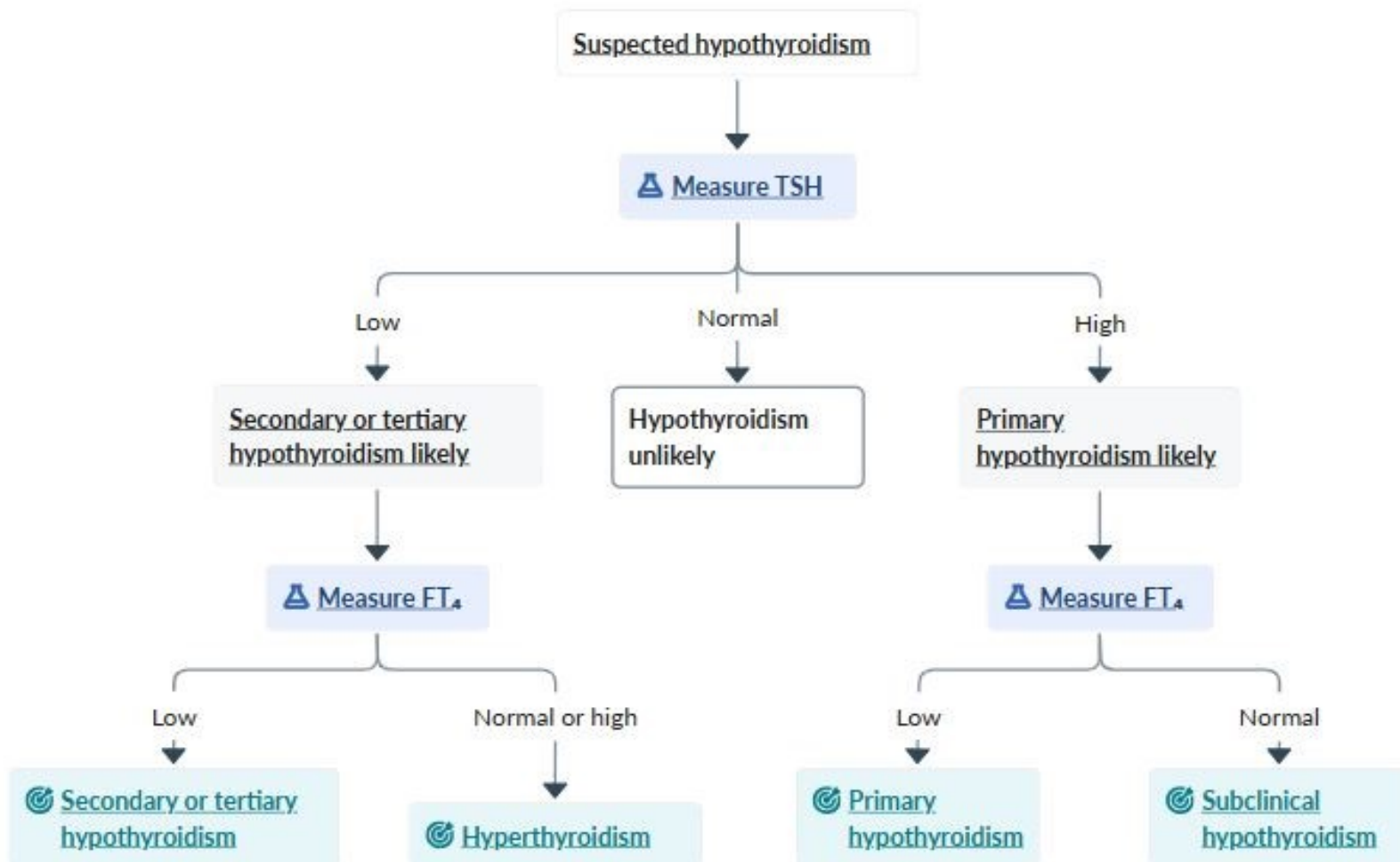


TABLE 38-2 *Diagnosis and Management of Thyrotoxicosis*

Etiology	Mechanism	Thyroid Examination	Lab Results	Radioiodine Uptake	Treatment
Graves disease	Stimulatory anti-TSH receptor antibodies	Diffuse goiter	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ +Antithyroid Abs ↑ thyroglobulin	↑ uptake diffusely	ATDs RAI Surgery
Toxic multinodular goiter or solitary nodule	Activating gene mutations	Nodule(s), usually >3 cm	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ -TSl/TRAb ↑ thyroglobulin	↑ focal uptake	RAI is the treatment of choice ATDs as temporizing measure Surgery if compressive symptoms or risk of thyroid cancer
Thyroiditis (thyrotoxic phase)	Follicular destruction causing release of stored hormone	Tender goiter in subacute thyroiditis; nontender goiter in "silent" thyroiditis	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↓ uptake	Prednisone
Iodine-induced	Iodine surplus (IV dye or amiodarone-induced thyrotoxicosis type 1)	Nodular or diffuse goiter	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↓ thyroglobulin	↓ uptake	Withdrawal of offending agent if possible. ATDs, RAI, or surgery in selected cases
Choriocarcinoma	Secretes hCG, a thyroid stimulator	Minimal goiter	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↑ uptake caused by increased thyroid hormone production	Treatment of choriocarcinoma
Pituitary tumor (TSH-secreting)	TSH overproduction	Goiter	TSH normal/↑ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↑ uptake	Treatment of pituitary adenoma
Struma ovarii	Ectopic thyroid tissue in ovarian tumor	Normal	TSH ↓ Normal/↑ FT ₄ Normal/↑ T ₃ - Antithyroid Abs ↑ thyroglobulin	↓ uptake in the region of the thyroid; ↑ uptake in the region of the pelvis	Treatment of struma ovarii
Exogenous thyrotoxicosis		Normal	TSH ↓ ↑ FT ₄ (if exogenous T ₄ intake) ↑ T ₃ (if exogenous T ₃ intake) - Antithyroid Abs ↓ thyroglobulin	↓ caused by suppression of thyroid hormone production	Withdraw offending agent
Iatrogenic or factitious	Over supplementation with thyroxine	Normal	TSH ↓ ↑ FT ₄ ↑ T ₃ - Antithyroid Abs ↓ thyroglobulin	↓ uptake	Withdraw offending agent
Drug-induced	Use of amiodarone, interferon-α, tyrosine kinase inhibitors	Normal	TSH ↓ ↑ FT ₄ ↑ T ₃ - Antithyroid Abs ↓ thyroglobulin	↓ uptake	Withdrawal of offending agent

Abs, Antibodies; ATDs, antithyroid drugs; IV, intravenous; RAI, radioactive iodine.

Case scenario :

A 50-year-old woman comes to the physician because of intermittent pain and numbness in her right hand for 6 weeks. She has a pins-and-needles sensation that worsens at night and is relieved when she shakes her hand. She also has episodic left knee pain throughout the day. She has a history of hypertension controlled with lisinopril. She takes over-the-counter medications for constipation. Her BMI is 35 kg/m². Her mother has a history of rheumatoid arthritis. She looks fatigued. Her pulse is 57/min and blood pressure is 120/75 mm Hg. On physical examination, there is normal range of motion in the wrists and digits. Sensation is decreased to light touch in the thumb and index finger. There is no thenar muscle atrophy. Deep tendon reflexes are 1+ and there is mild edema in the legs.

Which of the following treatments is most likely to benefit the patient?

- A. L-thyroxine
- B. Methotrexate
- C. Ibuprofen
- D. Surgical decompression
- E. Oral prednisone

Summary

I just need
the main ideas



References

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Thank You